

6. Scralgebra

The last application is a game of Scrabble with mathematical equalities that the player must construct. To launch the game, the user must click on the corresponding icon on the home screen. He will then be invited to turn his smartphone from 90 degrees to play the game.

6.1. Title Screen



This is the first screen the user will see when launching the application. He has four options. If he clicks on the “Play” button, a game will be launched (see [Game Screen](#)). If he clicks on the “Options” button, he will be able to configure the game settings (see [Options Screen](#)). If he clicks on the “Home” button, he will go back to the [home screen](#) of

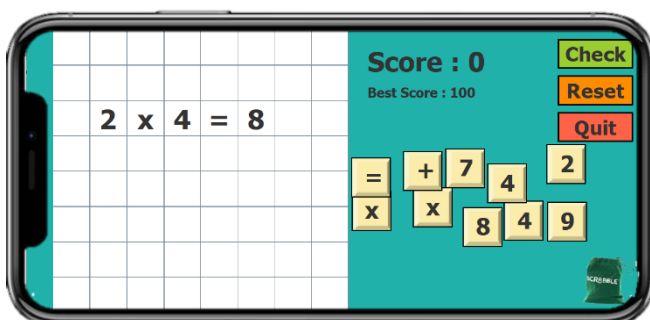
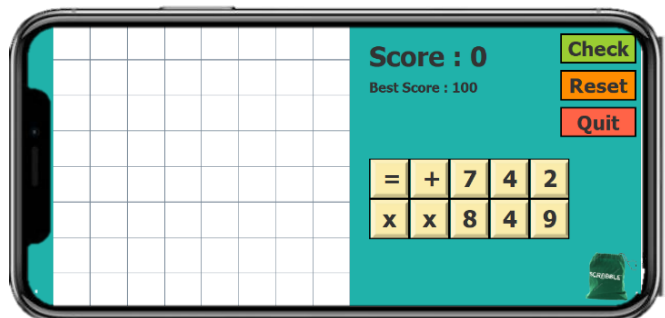
the smartphone. Finally, should he click on the “Credits button”, some information about the creation of the application will be displayed. The user can click on the same button again to go back to the original screen.

6.2. Game Screen



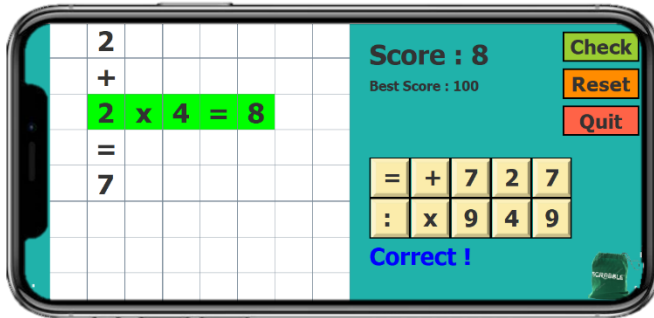
This is the screen where the game will be played. We will now explain the different steps the player must go through during a game.

First, he must click on the bag in the bottom right corner to draw new pieces.



He then must write a mathematical equality on the board on the left, using the drawn pieces. (He can move the pieces with the mouse if it helps him.) For example, he can here write “2x4=8”.

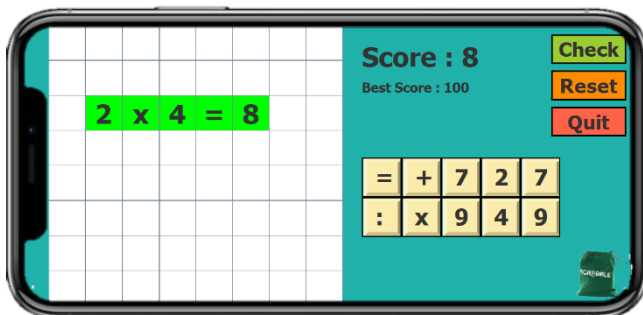
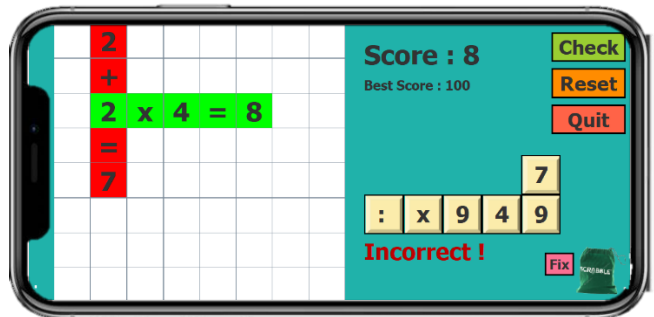
Once it is done, he clicks on the “Check” button. If he entered a correct equality, it will be highlighted in green, and used pieces will disappear. His score will be incremented by the value of the equality (here, 8 points).



The player then repeats the process : he draws new pieces and tries to write a new equality. He can also take advantage of the elements already on the board, like in the real Scrabble.



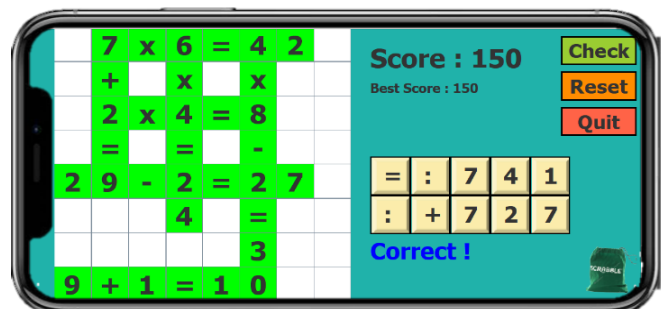
However, if the player tries to validate a wrong equality, or if he tries to use a piece that he did not draw and that was not already on the board, the score will not be incremented, an error message will be displayed and wrong elements on the board will be highlighted in red. If the player wants to quickly comeback to the state just before the validation, he can click on the “Fix” button.



The player can then continue. If he is lucky, he might receive a white piece (with no character on it). This piece works like in the real Scrabble : the player can use it to write any value for one character of his equality.



The goal of the game is to have the biggest score as possible. The highest score the player has achieved is displayed just below his score. If the player wants to restart the game, he can click on the “Reset” button. If he wants to return to the [title screen](#), he can click on the “Quit” button.



6.3. Options Screen



In this last screen, the player can adjust the game settings. We will now explain the meaning of each row.

- **Difficulty** : if the player does not want to waste time on the options, he can simply click on the difficulty level he wants, and every other parameters will then be automatically defined for him.
- **Timer** : the player can choose the quantity of time he will have at his disposal when playing a game. If he chooses the “Off” options, there will be no timer. Otherwise, a clock will be displayed on the game screen. If the timer reaches zero, the game is over. 
- **Lives** : the player can choose the number of lives he has when playing a game. If he chooses ∞ , he will have an infinite number of lives. Otherwise, a few hearts equal to the number of lives left will be displayed on the game screen. The player loses a life every time he makes a mistake. If he loses all his lives, the game is over. 
- **Operators** : the player can choose which operators may appear during a game. If he chooses “All”, the operators will be plus, minus, times, divided and the “factorial” (unadvised for someone with mathematical knowledge of low level).
- **Board size** : the player can choose the number of rows there will be on the squared board. The bigger it is, the more space he will have to form the equalities.
- **Number of Pieces** : the player can choose the maximum number of pieces that he can draw each turn. If ∞ is selected, there will be no pieces in the bag and the player can write any numbers or operators on the board.
- **Background Color** : the player can choose the background color of the application from four different types of green.

In addition to these settings, the player can also fill the checkbox “Sound” : in such a case, the musics, sound effects and voice clips of the application will be enabled. The information of the music playing will be displayed in the bottom of the game screen. Once the player has finished configuring the settings, he has 2 options. If he clicks on the “Cancel” button, his changes are not saved, and he will go back to the [title screen](#). If he clicks on the “Confirm” button, his changes will be saved, and a game will directly be launched with the new settings.